

The Field-Tensor Dictionary: From the Mechanical State of a Tensioned Two-Strand Rope to Electromagnetism

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July 2026. The rigorous assembly of the corpus’s electromagnetic identifications into a single mapping: mechanical state → potentials → field tensor → forces, with every line either Derived or explicitly flagged, and both force-sign rules obtained from mechanics with nothing put in by hand.

1. The mechanical state variables

The two-strand rope under tension T_0 carries one internal gapless degree of freedom: the braid phase Φ , the single Goldstone mode of the internal symmetry (the screw generator; the one-generator theorem, GRV-020). Its derivatives are the physically meaningful densities: the TWIST DENSITY $\eta = \partial\Phi/\partial s$ (structural handedness per unit length) and the ROTATION RATE $\omega = \partial\Phi/\partial t$ (the helix turning like a screw). The tension channel couples to twist through the twist-stretch lock, $k_{tw} = k + \lambda \gamma^2 \tau_0^2$ with $\gamma = 1/\sin^2\theta$ derived from helix geometry (EM-RECON-012). Coarse-graining the strand orientations over many ropes yields the orientation field, carrying a connection in the fiber-bundle sense (GG-001).

2. The dictionary

Electromagnetic object	Mechanical identity	Status	Claims
Charge q	Winding / linking number of the two strands: an integer; handedness is the sign	Derived (quantized)	GG-006, EM-001
Current J	Transported linking: the helix turned like a screw; one turn = one linking unit past each cross-section; $\text{div } J = 0$ is the no-wind-up condition	Derived	EM-008, EM-014
Vector potential A	The coarse-grained orientation connection; gauge freedom = phase-convention arbitrariness (physically: rigid braid-phase relabeling, the GRV-020 symmetry)	Derived (framework)	GG-001, GG-004

Scalar potential φ	The twist-tension channel: stored twist transmitted to tension by the twist-stretch lock	Modeled	EM-RECON-011, -012
$E_i = F_{i0} = -\partial_i \varphi - \partial_t A_i$	Static part: along-rope tension/twist gradient. Induction part: time-varying orientation transport (Faraday, inside $dF = 0$)	Assembled here	EM-RECON-011, EM-003
$B_i = \frac{1}{2} \epsilon_{ijk} F_{jk}$	Curl of the orientation field: the surrounding network's circulating response to the rotation rate ω	Derived / Modeled	EM-009, EM-010, EM-014
Field magnitude	$g = E \sqrt{(\epsilon_0 / \Sigma)}$: strain per field, with ONE constant $\Sigma = T_{on_L}$ (vacuum tension density), bounded by ATLAS/PVLAS	Modeled (calibrated form)	EM-RECON-014, -016
Maxwell equations	$dF = 0$ (Bianchi identity) + $d * F = J$ (Chern-Weil, $d = 3$)	Derived	EM-003
Lorentz force $q(E + v \times B)$	From the coupling $q \cdot v \cdot a$, FORCED by gauge invariance + current conservation; correct coefficient; $\perp$ to v and B ; does no work	Derived	EM-013, EM-010
Magnetic force sign	Gap geometry: same-sense circulations produce opposite swing velocities between wires $\rightarrow$ mesh $\rightarrow$ attract; opposite $\rightarrow$ clash $\rightarrow$ repel	Derived	EM-012
Electrostatic force sign	Constraint-source superposition: like twist fields ADD between knots (repel); opposite CANCEL (attract); $+q_1 q_2 / (4\pi K d)$	Derived (new)	EM-015

3. The electrostatic sign theorem (the new result)

The trap, named first: a quadratic field between sources does NOT automatically repel like charges. With COUPLING sources (an energy term $-\rho\Phi$, the structure of scalar exchange), the minimized interaction is $-q_1q_2/(4\pi Kd)$: like charges attract — which is why scalar-mediated forces attract and why naive arguments get Coulomb backwards. The rope winding is not a coupling source. Winding number is TOPOLOGY: a boundary condition on the braid-phase configuration space, with nothing to couple and nothing to switch off. For such CONSTRAINT sources the energy is purely $(K/2)|\nabla\Phi|^2$ on the constrained minimizer, and superposition gives $U = +q_1q_2/(4\pi Kd)$: like windings repel, opposite attract, with the $1/d$ potential and $1/d^2$ force of Coulomb's law. The laboratory precedent is exact: superfluid vortices are constraint-sourced quadratic fields, and like-sign vortices repel while vortex–antivortex pairs attract. Electron and positron are opposite handedness (GG-006): opposite winding, hence attraction. The benchmark verifies both the analytic dichotomy (the two source types provably carry opposite signs, so the trap is permanently fenced) and the mechanism on a lattice of actual winding defects: the like pair's energy falls with separation (repulsion), the opposite pair's rises (attraction). In plain terms: like-handed twist fields add in the region between the knots, storing more energy when they are close, so they are pushed apart; opposite-handed fields cancel between the knots, so they are pulled together. With the magnetic half (the gap-geometry rule of EM-012), the sign ledger of electromagnetism is now closed entirely by geometry.

4. The status ledger: the four debts, all discharged

The dictionary was assembled with four debts explicit. As of 2026-08-11 all four are discharged by derivation. The history is kept here rather than erased, because the route matters as much as the result.

(iv) UNIQUENESS - DISCHARGED (EM-017, Derived). Any mechanical assignment predicting the same force on every test winding at every velocity yields the same (E, B) pointwise: E follows from a static probe under the registered operational definition (force per winding), B by explicit inversion of $\mathbf{v} \times \mathbf{B}$ on three independent velocities, $\mathbf{F}_{\text{mu}}$ pointwise, and the potentials up to gauge by the Poincare lemma on EM-003's $d\mathbf{F} = 0$, gauge freedom being registered as physical phase-convention arbitrariness. The duality rotation, named in advance as the defeater, fails both arithmetically and structurally: charge is winding and no magnetic monopole is registered, so the rotated sources cannot be represented.

(iii) THE SCALAR POTENTIAL'S CHANNEL - DISCHARGED (EM-018, Derived). Over the closed inventory of registered mechanical channels, the transverse carrier is excluded by Helmholtz - every transverse mode obeys $\mathbf{k} \cdot \mathbf{e} = 0$ and is therefore divergence-free, while the curl-free and divergence-free subspaces meet only at zero, so a gradient E has exactly zero projection onto it. The screw/torsion mode is excluded structurally, since it carries winding and therefore IS the charge rather than its mediator. The longitudinal/tension channel is the only survivor, and its gaplessness - derived in EM-RECON-012 for the independent reason that u is gauge so no mass term is writable - is precisely what long-range Coulomb requires, a gapped mediator giving Yukawa rather than $1/r$.

(ii) THE INERTIAL TERM - DISCHARGED AS TO FORM (EM-019, Derived). The easy Newtonian move is unavailable here, since FND-REL-002 establishes there is no material velocity. What

works instead is that the same shift symmetry which forbids a mass term permits the inertial one, because a gauge label's rate of change is observable even when the label is not. Under $a \rightarrow a + c_0$ the mass term changes by $c_0(c_0 + 2a)/2$ and the mixed term by $c_0 da/dt$, while $(da/dt)^2$ and $(\text{grad } a)^2$ are exactly invariant. At two-derivative order only EM-007's stiffness and the inertial term survive. Residuals recorded: the coefficient is matched to EM-RECON-025's registered branch rather than independently derived, and the uniqueness argument assumes locality.

(i) SIGMA'S ABSOLUTE VALUE - DISCHARGED (EM-021). Sigma is not a free constant: it is its own definition $T_0 n_L$ evaluated on the registered mesh, where $n_L = 1/a^2$ per strand family, giving $6.0e36$ to $9.0e37 \text{ J/m}^3$ across the packing floors. The $\sim 1e25 \text{ J/m}^3$ formerly quoted here was never a measurement but a LOWER BOUND saturated by an assumed equality ($\text{Sigma} \geq \epsilon_0 E_S^2/g^{*2}$ from Schwinger-scale onset, with 'equality = the identification' stated in the source), and QGATE-007 withdrew the saturating assumption by showing that unpolarized light-by-light carries no structure information and that its rate is degenerate with normalization. A value eleven orders above a lower bound satisfies it: every combination of packing floor, n_L convention and g^* clears the Schwinger requirement by five to six orders. FND-031's independently registered $E_{\text{crit}} = 2.0e23 \text{ V/m}$ had already run this confrontation on the pinned Sigma.

Consequences, and the grade. Every identification in the dictionary is now forced: nothing in the mapping from mechanical state to field tensor is a choice, and the corpus's central identification claim GG-005 ('F IS the electromagnetic field; topology IS charge') no longer rests on any undischarged debt. On 2026-08-11 the author raised this assembly to DERIVED, registered as its own auditable act (EM-022) rather than folded into the discharges, because three of the four discharges were produced in the same session and a grade raised by its own evidence-producing session cannot otherwise be checked.

THE GRADE CARRIES THREE CONDITIONALITIES, which travel with it and are not shed. (C1) $\text{Sigma} = T_0 n_L$ evaluates on the registered mesh, which sits on the κ_{pack} floors held by FND-037's Conjecture-grade nonlinearity form; Sigma therefore spans $6.0e36$ to $9.0e37 \text{ J/m}^3$ across the floors. This is a parameter conditionality inherited from the inputs rather than an unfilled step in the mapping - as every claim inherits the electron-mass calibration - but it does mean a Derived grade resting on a chain whose upper reaches are Conjecture-held, and the paper does not pretend otherwise. (C2) The inertial term's coefficient is matched to EM-RECON-025's registered branch rather than independently derived. (C3) The inertial term's uniqueness argument assumes locality, so a genuinely nonlocal kernel would evade it.

A FALSIFIER IS ARMED, because a grade that cannot be lost is not a grade. If the κ_{pack} floors move materially, or if FND-037's quadratic-nonlinearity form is refuted, this assembly's grade returns to adjudication rather than keeping the label by seniority.

A downstream correction belongs with this section. EM-RECON-015's effective vacuum mass density of $\sim 1e8 \text{ kg/m}^3$ was computed from the superseded $\sim 1e25$ Sigma and is stale by ten to twelve orders. The live figures are $4.0e18 \text{ kg/m}^3$ on $\text{Sigma}_{\text{eff}}$ - matching FND-031's independently registered fence of $17.5x$ nuclear - and $2.0e20$ on the $3T_0/a^2$ reading. The

vacuum-energy problem that claim flagged is therefore substantially larger than it states, not smaller.

5. What would strengthen each line

Deriving Σ from the mesh scale and tension (blocked at present by the same irreducibility results that block a and $\hbar$); deriving EM-010's inertial term from strand dynamics; promoting the φ -channel identification by an independent prediction in the static sector; and a uniqueness argument — or a counterexample assignment, which would be equally valuable and would go into the registry the same way everything else has.

Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

6. The connection fixed: A is the rotated collective displacement (E-RECON, 5 September 2026)

The gap that surfaced. The dictionary's Faraday half, $E = -\text{grad } \varphi - dA/dt$ with A the orientation connection (EM-016), and the Magnus force identification $E = \rho\kappa_0(v_{\text{medium}} \times \hat{z})$ (EM-RECON-026) were both Derived-class statements about the same field, and until a boundary question was put to them nobody had checked that they agree. Commission CASIMIR-PLATE asked what a perfect conductor (tangential $E = 0$) imposes on the medium and found that, read as 'the strand slope', the connection gives a Neumann-class condition on the collective transverse displacement while the Magnus form gives a Dirichlet-class one; worse, for a transverse plane wave the two readings did not even agree on the polarization of E . The commission halted at its reading leg (CAS-BC-OPEN), as its charter required.

The reconciliation. Commission E-RECON (analysis/ERECON_results.md; EM-023, registered) fixed two candidate identifications of A in advance, the tilt ds/dz and the rotated displacement, and tested each symbolically against four conditions: pointwise agreement of $-dA/dt$ with the Magnus E for every medium motion, transverse E for a transverse plane wave, GG-004's gauge content, and reproduction of the Magnus form's constant $B' = -\rho\kappa_0\hat{z}$ as the background of curl A . Exactly one candidate passes all four: $A = -\rho\kappa_0(s \times \hat{z})$, with s the collective transverse displacement of EM-RECON-025. Under it $-dA/dt = \rho\kappa_0(v \times \hat{z})$ identically; a transverse wave along z polarized in x gives $E = \rho\kappa_0 f' \hat{y}$, in-plane and transverse; a uniform translation of s shifts A by the gradient of a linear function, pure gauge; and curl $A = -\kappa_0\rho \partial s/\partial z$ in-plane plus $\rho\kappa_0 \text{div}_\perp s$ along z , which is exactly $B = -\kappa_0 \rho_{\text{total}} \hat{z}$ with $\rho_{\text{total}} = \rho(1 - \text{div}_\perp s)$: the Magnus B' is the unperturbed background, its perturbation the mesh compression, and the in-plane part the strand tilt. B is minus κ_0 times the strand-direction flux density, which makes literal the phrase this dictionary already used for the static field, 'rest-tension flux geometry, rope count' (EM-RECON-011).

What changes in the table above. The row for the vector potential keeps its status; its mechanical identity is now stated precisely: the orientation connection is the rotated collective displacement, and 'orientation-transport rate' in the E row reads as the rate of the transported displacement, not the strand slope. The tilt candidate gives a different field for the same motion and is excluded. EM-016's grade is untouched; a reading it left open is fixed, consistent with EM-017's operational uniqueness. The corollary the boundary question was waiting for follows uniquely: a perfect conductor pins the tangential collective displacement at the plate (Dirichlet on both transverse bands, effective separation $d + 1$ on the lattice) and levels the twist-tension field ϕ along it. The Casimir computation this unblocked is in progress; its result belongs to a later revision of this paper.

Registered Claims in This Paper

Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **EM-015 [Derived]:** ELECTROSTATIC SIGN THEOREM [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-015 [Derived]:** ELECTROSTATIC SIGN THEOREM [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-016 [Modeled]:** THE FIELD-TENSOR DICTIONARY [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-016 [Modeled]:** THE FIELD-TENSOR DICTIONARY [benchmark: benchmarks/em/electrostatic_sign_derivation.py]
- **EM-023 [registered]:** THE ORIENTATION CONNECTION IS THE ROTATED COLLECTIVE DISPLACEMENT, $A = -\rho\kappa_0 (s \times \hat{z})$; EM-016's E and EM-RECON-026's E are one field; a perfect conductor pins the tangential displacement (analysis/ERECON_results.md)